

LORENZO SPIRIDIONI

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Summary

Cloud Backend Engineer focused on Kubernetes-based distributed systems, multi-cloud infrastructure (AWS, Azure), and LLM/ML pipeline deployment. Research background in RAG-based clinical applications and microservice observability at scale.

Experience

Rox Pay

Jan. 2025 – Present

Software Engineer

Remote, Italy

- Designed a distributed backend serving **1KK+ messages/hour** in **C# .NET 9** on **Kubernetes** with **RabbitMQ**, **Redis**, and **Elasticsearch**.
- Built and operated ML inference microservices (**Python/FastAPI**, **LangChain**, **PostgreSQL**) achieving **99.5% availability** in production.

CCH Tagetik — Wolters Kluwer

Mar. 2023 – Jan. 2025

Cloud Engineer

Lucca, Italy

- Authored **30+ Terraform modules** and **Helm charts** for multi-region **AWS/Azure**, deployed via **Jenkins** pipelines **reducing infrastructure costs by 40%** through standardized configuration management.
- Automated configuration of **1000+ servers** with **10+ different OS/distro** with **Chef/Ansible**; containerized **Python** services on **Docker/Kubernetes**, reducing build time by **80%**.

Amazon

May 2022 – Sept. 2022

IT Support Engineer Intern

Civitate Al Piano, Italy

- Built a **Python**-based network discovery and topology mapping system covering **15,000+ devices**, reducing manual documentation effort by **95%** and saving an estimated **20 engineering hours/week**.

Education

Master in Computer Science – Big Data Technologies

Sep. 2023 – Present

University of Pisa — Freie Universität Berlin

Pisa, Italy — Berlin, Germany

- Coursework: Machine Learning, Deep Learning, Large-Scale Data Analytics, Cloud Computing. Capstone: LLM chatbot for **Charité Hospital** (see Projects).

Bachelor in Computer Science

Sep. 2020 – Feb. 2023

University of Pisa

Pisa, Italy

- Completed in 2 years** instead of 3 (top 5%). Thesis: Novel error-tracing strategy for **Istio** service mesh, in collaboration with **Italian National Research Council**, reducing root cause identification time by 66%.

Projects

LLM Chatbot for Charité University Hospital | *Python, FastAPI, LangChain, LangGraph, Docker, PostgreSQL*

- Led a team of 20 to build a **RAG-based clinical chatbot** for Charité Hospital using **LangChain/LangGraph**, indexing **4,200+ medical documents** into a **Qdrant** vector store for semantic retrieval. Presented live to **12 physicians** at Freie Universität Berlin, with incoming publication.
- Deployed via **Docker Compose**: FastAPI backend, React frontend, PostgreSQL and Qdrant RAG database, achieving **<1 min** cold-start.

Anomaly Detection Platform | *Kubernetes, Prophet, Prometheus, Python*

- Deployed a **Prophet**-based forecasting microservice on **AWS EKS**, processing **50K+ data points/day** across 4 endpoints. Automated **Grafana** alerting, reducing mean time to detect and **reducing resources costs by 15%** through fine-grained CPU/memory limit tuning.

Technical Skills

Languages: Python, C#, Rust

ML & AI: LangChain, LangGraph, PyTorch, Prophet, RAG Pipelines, Prompt Engineering

Cloud & DevOps: AWS, Azure, Terraform, Ansible, Chef, Jenkins, ArgoCD, Helm

Containers & Orchestration: Kubernetes, Docker, Docker Compose

Data & Messaging: PostgreSQL, Redis, RabbitMQ, Elasticsearch

Observability: Prometheus, Grafana, Istio, Jaeger

Compliance & Standards: ISO 27001, ISO 9001